

Linear Algebra

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See the structure before calculating

Quick reference

$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$. For a triangular matrix, multiply the diagonal entries.

A zero row, or two proportional rows, makes the determinant zero. A zero *diagonal entry* alone is not enough.

Spot a reason, not a long calculation

For each matrix, decide whether its determinant is zero without cofactor expansion. Give a specific structural reason. Also give the value for C.

$$A = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 0 & 0 \\ 1 & 4 & -2 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 2 & -1 \\ 3 & 6 & -3 \\ 0 & 1 & 2 \end{pmatrix}$$

$$C = \begin{pmatrix} 2 & -1 & 4 \\ 0 & -3 & 1 \\ 0 & 0 & 5 \end{pmatrix}$$

Matrix	Zero / nonzero (value for C)	Structural reason
A		
B		
C		

A useful counterexample

Do zeros on the diagonal alone force a zero determinant? Use D to test that claim.

$$D = \begin{pmatrix} 0 & 2 \\ 3 & 0 \end{pmatrix}$$

Calculate $\det(D)$, then give a one-sentence conclusion

Check your understanding

What extra condition lets you use the diagonal-product rule? Explain the difference between a **zero row** and a **zero diagonal entry**.

Your explanation

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Track the effect of every row operation

Three operations, three determinant rules

Swap: $R_i \leftrightarrow R_j$ reverses the sign.

Scale: $R_i \leftarrow cR_i$ multiplies the determinant by c .

Replace: $R_i \leftarrow R_i + kR_j$ ($i \neq j$) leaves it unchanged.

Reduce only as far as you need

Find $\det M$ using row operations. Stop when the matrix becomes triangular. Record how each operation affects the determinant.

$$M = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 4 & 3 \end{pmatrix}$$

New matrix / row operation	Effect on the determinant

Determinant of your triangular matrix

Now recover $\det(M)$: account for every sign or scale change

Check your understanding

“Row operations never change the determinant.” Correct the sentence precisely. Why does your final value refer to the original matrix, not just the final one?

Your explanation

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Choose an efficient expansion; check it

Cofactor expansion

For expansion along row i : $\det A = \sum_j a_{ij}(-1)^{i+j} \det A_{ij}$.

A_{ij} is obtained by deleting row i and column j . Keep **entry**, **sign** and **minor** separate.

Use the zeros

Calculate $\det N$ by cofactor expansion. Choose an efficient row or column and write the sign and minor explicitly. Then check using a different row or column.

$$N = \begin{pmatrix} 2 & 0 & 3 \\ 1 & -2 & 0 \\ 4 & 0 & 1 \end{pmatrix}$$

First expansion: chosen row / column, terms and calculation

Independent expansion: a different row / column

Why must the two answers agree?

Reason

Check your understanding

$$T = \begin{pmatrix} 4 & 2 & -1 \\ 0 & -1 & 3 \\ 0 & 0 & 2 \end{pmatrix}$$

Start with T . Swap rows 1 and 3 to obtain S . Then apply $R_2 \leftarrow R_2 + 5R_1$ to S to obtain V .

$\det T =$ _____ $\det S =$ _____ $\det V =$ _____

Name the rule at each step
